AMITY UNIVERSITY
MADHYA PRADESH
Established vide Government of Madhya Pradesh Act No. 27 of 2010

Syllabus

Programme Name	B. Tech. (CSE)		BASIC	Batch	2025-2029
Course Code	CSC 601	Course Name	Web Application Security And Penetration Testing Design	Semester	VI

Credits (Total)	L	T	P	Marks (Internal/External		Contact Hours (per week)	Independent Study Hour (per week)
3	3	0	0	30	70	3	3

Course Objective:

The course will cover the concepts involved in web application development. The course will introduce to various vulnerabilities in web applications and their mitigation techniques.

Course Alignment with UNSDG:

The Course aims to fulfill the United Nations Sustainable Development Goals SDG 4 (Quality Education)

Course outcomes:

After completion of course, the student will be able to:

CO-1	Apply client-side web development to design interactive front-end web user interfaces.
CO-2	Understand server-side web application concepts to develop back-end web server application
CO-3	Identify and mitigate various client-side web application security vulnerabilities
CO-4	Identify and mitigate various server-side web application security vulnerabilities

Teaching Pedagogy:

T1	Activity based learning through lab experiments like Dissection, specimen observation Power Point Presentations and white board teaching
T2	Class Test, Assignments, Flip Class/ Seminars, Quiz, Oral Viva-voce examination

Assessment Tools

AT1-1	Quiz	**AT1-6**	Poster
AT1-2	Midterm Exams	**AT1-7**	Oral Viva-voce examination
AT1-3	Flip Class	**AT1-8**	
AT1-4	Seminar Presentation	**AT1-9**	
AT1-5	Assignments		

Prerequisites: Basics of Cyber security and Computer Networking

Module	Descriptors/Topics	Hours	Assessment tools
I	**Introduction:** Web application development – Introduction - Architecture – Client-side technologies and frameworks – HTML – CSS – Java Script - Ajax/Fetch - Data interchange formats – XML, JSON. Sever-side scripting and technologies - development – technologies - Handling client requests – Database connectivity – Sessions – Cookies	9	Quiz Mid-term Exam Assignment
II	**Security threats and Vulnerabilities:** Web application vulnerabilities – Client-side Vulnerabilities - Cross Site Scripting (XSS) - Cross Site Request Forgery (CSRF) - Cross-origin resource sharing (CORS) – Clickjacking; Server-side Vulnerabilities - SQL injection - OS command injection - Directory traversal - Authentication - Server-side request forgery (SSRF).	9	Mid-Term Quiz Assignment
III	**Protocol engineering:** Experiments with open-source network simulators (NS2 and NS3): Installation and configuration, Creation of network topology and understanding of packet switched network, Simulation and visualization of different types of traffic—congestion controlled, and non-congestion controlled, Trace analysis and visualization of protocol dynamics.	9	Mid-Term Oral Viva- voce examination Seminar Presentation
IV	**Implementing network security Protocols:** Simulation with active queue management schemes, Simulation and visualization of attacks (e.g. IP spoofing and reflection attacks). Socket programming: implementation of IP spoofing and reflection DDoS attacks. Linux Kernel: Familiarization with Linux kernel protocol implementation (TCP/IP) implementation, Tracing and debugging of Linux Kernel TCP/IP source code, Kernel modification and recompilation, Implementation of a non- attack in Linux kernel Network Emulation and testbeds: Network emulation and traffic control using tc (traffic control), dummynet and other advanced tools, Familiarization with advanced testbed technologies (e.g. Emulab, DETER and PlanetLab, etc.), formal verification of protocol: SPIN, UPPAL.	9	Quiz Assignment Industrial Visit Report Seminar Presentation
V	**Design and Implementation of Penetration Testing:** Causes of Vulnerabilities, Penetration Testing Tools, Use external penetration testers to find problems, Planning and reconnaissance, scanning of ports, Penetration testing methods, Feedback results to the defect management and mitigation system, Scheduling of periodic penetration tests for application coverage, use external penetration testers to perform deep-dive analysis, Customize penetration testing tools.	9	Quiz Assignment Industrial Visit Report Seminar Presentation
	Total Teaching Hours	45	

Additional Resources

A. Value addition to course content/ Skill enhancement content

a. https://www.cybrary.it/course/cyber-security-for-beginners/

B. Remedial classes for slow learners

As per the AUMP SOP for Slow & Advanced Learn

Suggested reading:

1. Textbooks:

- W. Stallings, Network Security Essentials: Applications and Standards,6th Edition,Pearson Prentice Hall, 2016.
- Bryan Sullivan and Vincent Liu, Web Application Security, A Beginner's Guide, McGraw-Hill Education, 2012.
- Kaufman, R. Perlman and M. Speciner, Network Security: Private Communicationin a Public World, 2nd Edition, Prentice Hall PTR, 2002

2. Reference books

- Internet of Things: A Hands-On Approach by Arshdeep Bahga, Vijay Madisetti

3. Suggested e- resources (Websites/e-books)

- https://attack.mitre.org/

Assessment Plan:

Component of Evaluation	**Description**	**Code**	**Weightage %**
Continuous Internal Evaluation	Mid Term	CT	15%
	Home Assignment/Seminar/ Viva-Voce/Quiz	S/ HA/ V/Q	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves. including medical leaves.	A	5%
End Semester Examination	End Semester Examination	ESE	70%
Total			**100%**

Abbreviations: CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, ESE: End Semester Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

Course Outcomes	Correlation with POs												Correlation with PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	1	2	3						2	3	2	2	
CO2	1	1	1											2	
CO3	1	1	1	3							2		2	1	
CO4	1	**2**	1	3									2	1	

1: weakly related, 2: moderately related and 3: strongly related